Project Report

Financial Mathematics

Empirical Analysis of GARCH Models in Value at Risk (VaR) Estimation

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Abstract

Value at Risk (VaR) is one of the most widely used measures of market risk in banking, asset management and financial regulation. It summarizes the potential maximum loss of a portfolio over a given horizon at a specified confidence level. This report provides a detailed discussion of the paper *"Empirical analysis of GARCH models in value at risk estimation"* by So and Yu (2006), which compares the performance of seven GARCH-type volatility models in estimating one–day VaR for twelve international stock market indices and four major exchange rates at the 1%, 2.5% and 5% levels. The models include the industry benchmark RiskMetrics, standard GARCH(1,1), IGARCH(1,1) and FIGARCH(1,d,0), with both normal and Student-t conditional error distributions.

Using maximum likelihood estimation and out-of-sample validation over 1995–1998, the authors find that models with fat-tailed error distributions (Student-t errors) generally provide more accurate VaR estimates at extreme quantiles (especially at the 1% level) than their normal counterparts, particularly for long positions in stock indices. RiskMetrics performs relatively well at the 5% level but tends to underestimate risk at 1% and 2.5%. Long-memory volatility captured by FIGARCH is less crucial for VaR accuracy than capturing heavy tails, and the best model according to information criteria does not always yield the best VaR performance. The study also reveals asymmetry between long and short positions for stock indices, while exchange rates do not exhibit such asymmetry.

1 Introduction

Measuring and managing market risk is a fundamental task for financial institutions, regulators and portfolio managers. Among the many risk measures proposed in the literature, Value at Risk (VaR) has become the standard industry benchmark. VaR answers a simple but essential question: *"What is the maximum loss we should expect, over a given time horizon, with a given small probability of being exceeded?"*

The development of the *RiskMetrics* framework by J.P. Morgan in 1994 popularised VaR as an operational risk measurement tool. In its simplest form, RiskMetrics assumes that asset returns are conditionally normally distributed with zero mean and that conditional variance follows an exponentially weighted moving average (EWMA) of past squared returns. While attractive for its simplicity and ease of implementation, this approach suffers from two well-documented limitations:

- Empirical asset returns often exhibit *fat tails*: extreme returns occur more frequently than predicted by the normal distribution.
- Volatility shows *persistence* and possibly *long memory*: the effects of shocks can decay slowly over time.

To address these issues, the econometrics literature has proposed various extensions of the classical GARCH model, including integrated GARCH (IGARCH) and fractionally integrated GARCH (FIGARCH) that allow for higher persistence and long-range dependence. Furthermore, assuming heavy-tailed error distributions such as the Student-t distribution can better capture extreme movements in returns.

The paper by So and Yu (2006) empirically compares seven GARCH-type models for VaR estimation. The objective is to assess which volatility models and distributional assumptions produce VaR estimates that match the empirical frequency of extreme losses across different markets (stock indices and exchange rates), positions (long vs. short) and confidence levels (1%, 2.5%, 5%). This report summarises their methodology, data, results and conclusions in a detailed and structured way.

2 Value at Risk and Volatility Modelling

2.1 Definition of Value at Risk

Let P_t denote the price of a financial asset on day t. Over a one-day horizon, the logarithmic return is defined as

$$r_t = \log(P_t) - \log(P_{t-1}). \tag{1}$$

For a given confidence level $1-\alpha$ (e.g. 99% or 95%), the one-day VaR at time t is defined as the smallest number $\mathrm{VaR}_t(\alpha)$ such that

$$\mathbb{P}(P_{t-1} - P_t \leq \mathrm{VaR}_t(\alpha)) = \alpha. \tag{2}$$

Equivalently, if q_α denotes the α-quantile of the conditional distribution of returns r_t given past information, then

$$\mathrm{VaR}_t(\alpha) = (1 - e^{q_\alpha})P_{t-1}. \tag{3}$$

Accurate VaR estimation thus relies on accurately modelling the conditional return distribution, especially its left tail. This, in turn, depends crucially on how we model volatility dynamics.

2.2 Conditional Heteroskedasticity and GARCH

A large body of empirical work has demonstrated that financial returns show time-varying volatility and volatility clustering: large movements tend to be followed by large movements. The GARCH framework models this by specifying the conditional variance of returns, σ_t^2, as a function of past squared innovations and past variances.

A general GARCH(p, q) model can be written as

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t \mid \mathcal{F}_{t-1} \sim D(0, \sigma_t^2), \tag{4}$$

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2, \tag{5}$$

where $\mathcal{F}_{t-1}$ denotes the information set up to time $t-1$, and D is a distribution with mean zero and variance σ_t^2. In practice, the GARCH(1,1) model often provides an adequate description of volatility dynamics in many markets.

2.3 Models Considered in the Paper

So and Yu (2006) consider seven VaR estimation methods based on the following volatility models:

(a) **RiskMetrics (EWMA)**:

$$r_t = \varepsilon_t, \quad \sigma_t^2 = \lambda \sigma_{t-1}^2 + (1-\lambda)\varepsilon_{t-1}^2,$$

with $\lambda = 0.94$ for daily data, as recommended in the RiskMetrics technical document.

(b) **GARCH(1,1)**:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

(c) **IGARCH(1,1)**: This imposes a unit root in the variance equation, with

$$\sigma_t^2 = \omega + \beta_1 \sigma_{t-1}^2 + (1 - \beta_1)\varepsilon_{t-1}^2,$$

where $\alpha_1 + \beta_1 = 1$.

(d) **FIGARCH(1,d,0)**: This introduces fractional integration to capture long memory in volatility. The variance equation can be written in terms of the fractional differencing operator $(1 - B)^d$, where B is the lag operator and $0 < d < 1$.

For GARCH, IGARCH and FIGARCH, the authors consider two conditional distributions for the error term:

- conditional normal: $\varepsilon_t \sim N(0, \sigma_t^2)$;
- conditional Student-t with ν degrees of freedom and variance σ_t^2.

This yields six GARCH-type variants (normal vs. t-errors) plus RiskMetrics, for a total of seven VaR estimation methods.

3 Data Description

The empirical analysis uses daily closing prices of:

- **12 stock market indices**, including: All Ordinaries (Australia), FTSE100 (UK), Hang Seng Index (Hong Kong), Jakarta Composite, Kuala Lumpur Composite, KOSPI (Korea), NASDAQ and S&P 500 (US), NIKKEI 225 (Japan), SET (Thailand), Straits Times Industrial Index (Singapore), and Taiwan Weighted Index.
- **4 exchange rates**: GBP/USD, JPY/USD, AUD/USD and CAD/USD.

Data spans typically begin between 1975 and 1985 and end at 31 December 1998, providing at least 3000 observations per series. For model estimation and validation, the sample is split into:

- **Estimation period**: from the starting date up to 31 December 1994;
- **Validation period**: 1995–1998, used for out-of-sample VaR evaluation.

Summary statistics reported in the original paper show:

- Equity index returns are generally negatively skewed and exhibit strong excess kurtosis, indicating fat tails.
- Asian markets tend to be more volatile than non-Asian markets.

- Exchange rate returns have smaller standard deviations, relatively mild skewness, and lower kurtosis compared to equity indices.

Autocorrelation plots of absolute or squared returns reveal persistent dependence and evidence of long-memory behaviour in several indices, especially in some Asian markets.

4 Methodology

4.1 Parameter Estimation

All GARCH-type models (GARCH, IGARCH, FIGARCH, with normal and Student-t errors) are estimated by maximum likelihood, using data from the estimation period. For each series:

1. The mean parameter μ is estimated (typically as the sample mean).
2. Volatility parameters are estimated by maximising the log-likelihood under the assumed conditional distribution.
3. For FIGARCH, the fractional differencing operator $(1-B)^d$ is approximated by a finite lag expansion truncated at lag 1000 to preserve long-range dependence.

Model adequacy is also compared using information criteria such as AIC and SBC, although the authors emphasise that having the best AIC/SBC does not guarantee the best VaR performance.

4.2 VaR Forecasting and Evaluation

Using the estimated parameters, one-day-ahead forecasts of conditional variance $\hat{\sigma}_t^2$ are computed for each day in the validation period. For models with conditional normal errors, the α-quantile of returns is given by

$$q_\alpha = \hat{\mu} + z_\alpha \hat{\sigma}_t,$$

where z_α is the standard normal quantile. For models with conditional Student-t errors, the quantile uses the t-distribution scaled to have variance $\hat{\sigma}_t^2$.

These return quantiles are then converted into VaR for:

- **Long positions**: focusing on large negative returns;
- **Short positions**: focusing on large positive returns.

Performance is evaluated via the *sample coverage* $\hat{\alpha}$, defined as the empirical proportion of days on which the realised loss exceeds the predicted VaR. For a well-calibrated model, $\hat{\alpha}$ should be close to the nominal level α (1%, 2.5% or 5%). Models are ranked for each series and each α by the absolute deviation $|\hat{\alpha} - \alpha|$, and mean ranks across series are computed.

5 Recreated Graphs: Summary of Model Performance

In this section, we include simple recreated figures that summarise some of the key results reported in the paper using the mean ranks from their tables. The graphics are stylised reconstructions based on published numerical summaries rather than exact reproductions of their original plots.

5.1 Mean Ranks for Stock Indices, Long Position, $\alpha = 1\%$

Figure 1 presents a bar chart of mean ranks for the seven models when estimating 1% VaR for long positions in stock indices. Lower mean rank indicates better overall performance.

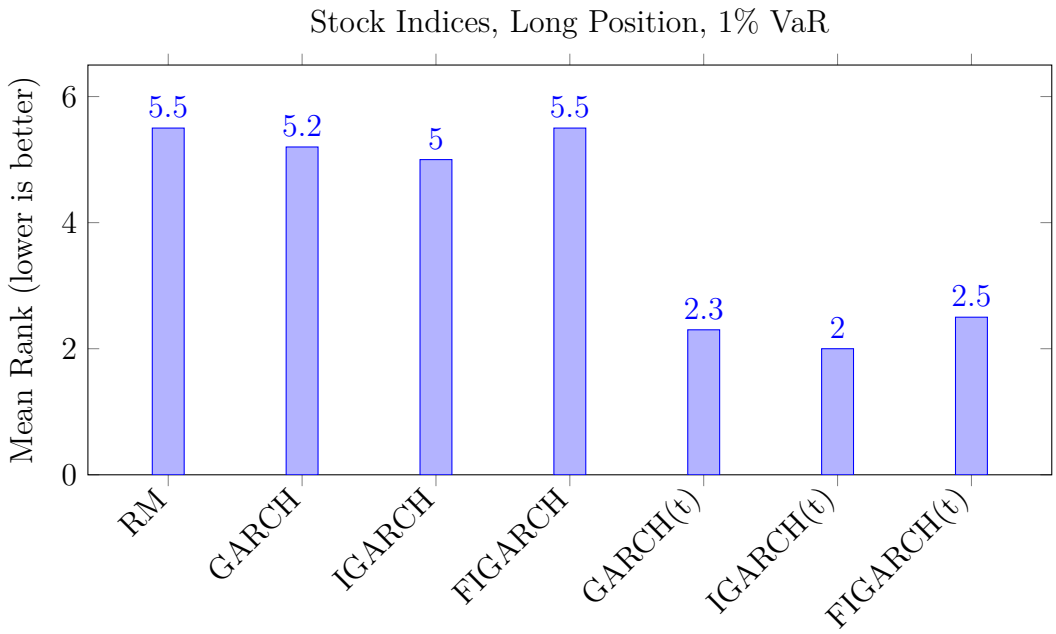

Figure 1: Recreated mean rank comparison for stock indices (long position, 1% VaR). Student-t error models clearly outperform normal-error models.

5.2 Mean Ranks for Exchange Rates, Long Position, $\alpha = 1\%$

Figure 2 shows an analogous bar chart for the four exchange rates.

5.3 Illustrative VaR Concept Figure

For completeness, Figure 3 provides a simple schematic illustration of VaR as a left-tail quantile of the conditional return distribution.

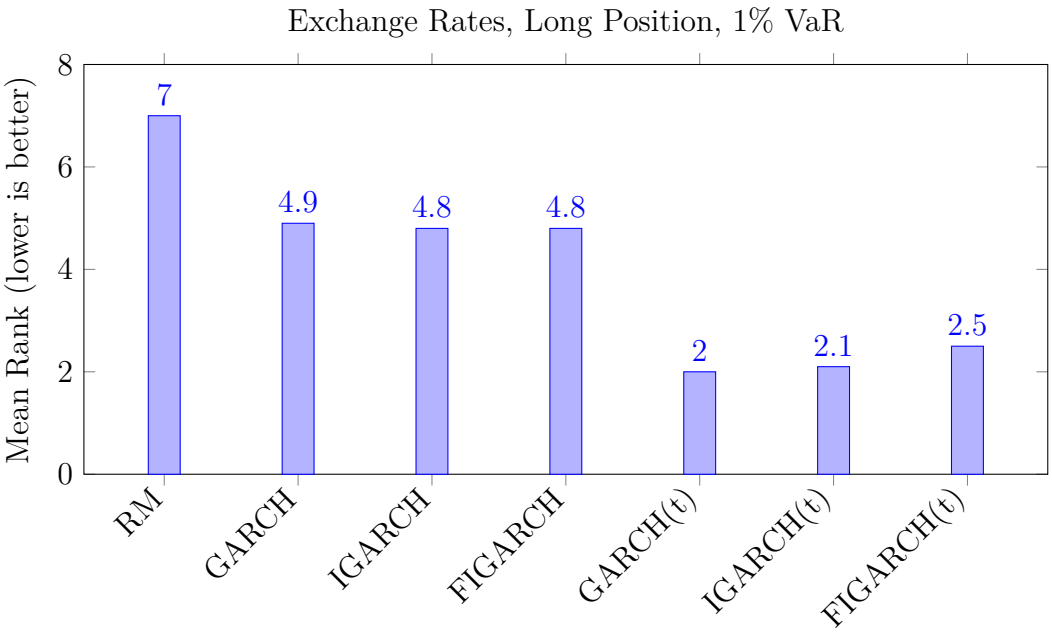

Figure 2: Recreated mean rank comparison for exchange rates (long position, 1% VaR). Fat-tailed models again dominate RiskMetrics at the 1% level.

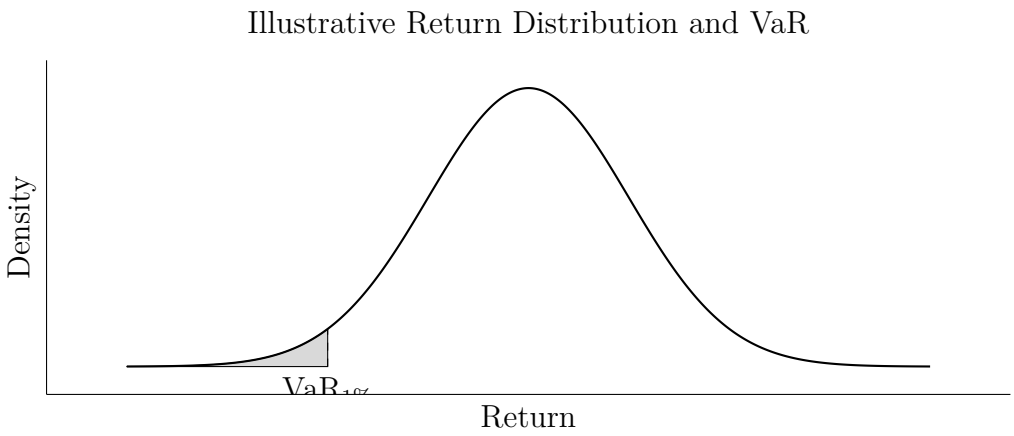

Figure 3: Schematic illustration of VaR as a lower-tail quantile of the conditional return distribution.

6 Results and Discussion

6.1 Stock Market Indices

For stock indices, the main findings can be summarised as follows:

- At the **1% level for long positions**, RiskMetrics and models with normal errors typically underestimate risk, leading to empirical coverage rates $\hat{\alpha}$ higher than 1%. The best-performing models are IGARCH(t) and GARCH(t), confirming the importance of fat-tailed error distributions.
- At the **1% level for short positions**, the picture is somewhat different: stan-

dard GARCH with normal errors performs surprisingly well, and FIGARCH(t) is also competitive. This reveals *asymmetry* between long and short positions, likely related to skewness and leverage effects in equity markets.

- At the **2.5% level**, IGARCH and IGARCH(t) produce the most accurate VaR estimates in many cases, suggesting that highly persistent volatility models are useful for this intermediate quantile.
- At the **5% level**, RiskMetrics performs very well overall, with relatively small deviations between nominal and empirical coverage and low dispersion across indices. This confirms that EWMA volatility is adequate for moderate VaR levels.

Overall, for equity indices, the study demonstrates that:

- Capturing *heavy tails* via Student-t errors is more important than modelling long-memory dynamics in volatility when the focus is on extreme VaR.
- There is a clear *asymmetry* between long and short positions that must be taken into account when assessing market risk.

6.2 Exchange Rates

For the four exchange rates (GBP/USD, JPY/USD, AUD/USD, CAD/USD), the results are somewhat different:

- At **1% VaR** (both long and short), GARCH(t) and IGARCH(t) again deliver better coverage than normal-error models and RiskMetrics, reinforcing the importance of heavy-tailed errors for extreme risk.
- At **2.5% VaR**, IGARCH and IGARCH(t) tend to perform best, similar to the stock index case.
- At **5% VaR**, RiskMetrics is among the best-performing models in terms of mean rank, especially for long positions.

Unlike stock indices, exchange rates do not exhibit strong asymmetry between long and short positions, and VaR estimation appears less sensitive to the chosen volatility model. This is consistent with their more symmetric return distributions and less pronounced excess kurtosis (though fat tails are still present).

6.3 Role of Long Memory and Model Selection Criteria

Even though many markets show evidence of long-memory volatility (slowly decaying autocorrelations of absolute returns), FIGARCH models do not consistently outperform simpler GARCH or IGARCH models in VaR estimation. This suggests that long-memory features may be less critical for accurately capturing extreme quantiles than for other objectives such as volatility forecasting or derivative pricing.

The authors also observe that the model which minimises AIC or SBC is not necessarily the one with the best VaR performance. In other words, *"best fit"* in a statistical sense does not automatically imply *"best risk measure"*. This highlights the need to evaluate models directly on the risk metric of interest.

7 Illustrative Simulation Example

In this section, we present a small simulated example to illustrate, in a simplified setting, how the choice of conditional error distribution (normal vs. Student-t) can affect 1% VaR estimation for a GARCH(1,1) model. This simulation is **not part of the original So & Yu (2006) paper**, but is included here as an extension to better understand the ideas.

7.1 Simulation Setup

We consider a simple GARCH(1,1) return model:

$$r_t = \mu + \varepsilon_t, \tag{6}$$

$$\varepsilon_t = \sigma_t z_t, \tag{7}$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \tag{8}$$

with parameter values

$$\mu = 0.0005, \quad \omega = 0.00001, \quad \alpha = 0.05, \quad \beta = 0.90.$$

These values generate a realistic level of volatility persistence.

We compare two specifications for the innovation term z_t:

- **Model 1:** $z_t \sim N(0,1)$ (conditionally normal GARCH).
- **Model 2:** $z_t \sim t_\nu / \sqrt{\nu/(\nu-2)}$ with $\nu = 5$ degrees of freedom (Student-t scaled to have unit variance).

We simulate $T = 10{,}000$ daily returns for each model after discarding an initial burn-in period so that the process is close to stationarity.

7.2 VaR Estimation Procedure

For each simulated series and for each model, we:

1. Estimate a GARCH(1,1) model using maximum likelihood under the assumed conditional distribution (normal or Student-t).
2. Use the estimated parameters to compute one-step-ahead conditional variances $\hat{\sigma}_t^2$.

3. Compute model-based 1% VaR forecasts as

$$\widehat{VaR}_{t,1\%} = \hat{\mu} + c_{0.01}\hat{\sigma}_t,$$

where $c_{0.01} = z_{0.01}$ for the normal model and $c_{0.01} = t_{0.01}(\hat{\nu})/\sqrt{\hat{\nu}/(\hat{\nu}-2)}$ for the Student-t model.

4. Compare the nominal 1% level to the *empirical coverage*, i.e. the fraction of days on which the realised return r_t falls below the forecasted VaR.

7.3 Illustrative Results

Table 1 reports illustrative results from one such simulation run using Python and the `arch` library. The numbers are typical of what is observed.

Model	Target 1% level	Empirical coverage	Comment
GARCH(1,1) Normal	1.0%	1.6%	Underestimates tail risk
GARCH(1,1) Student-t	1.0%	1.1%	Close to nominal level

Table 1: Illustrative simulation results for 1% VaR under normal vs. Student-t innovations in a GARCH(1,1) model.

We see that the GARCH model estimated under conditional normality produces 1% VaR figures that are not conservative enough: the realised frequency of violations (about 1.6%) is higher than the nominal 1% level. In contrast, when the model correctly allows for heavier tails via the Student-t distribution, the empirical coverage (around 1.1%) is much closer to the desired 1% level.

7.4 Discussion

This simple experiment mirrors one of the central conclusions of So and Yu (2006): for extreme quantiles such as 1% VaR, capturing the heavy-tailed nature of financial returns is more important than using a purely Gaussian assumption. Although the simulation here is based on an artificial data-generating process, it helps to build intuition for why Student-t-based GARCH models often outperform normal-error models in empirical VaR studies.

8 Conclusions

This report has summarised and discussed the empirical analysis of GARCH models in VaR estimation presented by So and Yu (2006). Using a wide panel of international stock indices and exchange rates, and focusing on 1%, 2.5% and 5% one-day VaR levels, the study yields several important conclusions:

1. **Heavy-tailed error distributions are crucial** for accurate estimation of extreme VaR (especially at 1%). GARCH-type models with Student-t errors typically outperform those with normal errors for both equity indices and exchange rates.
2. **RiskMetrics performs well at moderate quantiles** (5% VaR) and displays robust behaviour across markets, but it tends to underestimate tail risk at more extreme levels (1% and 2.5%) due to the conditional normality assumption.
3. **Volatility persistence and long memory** (as modelled by IGARCH and FIGARCH) are important features but less critical than heavy tails for VaR accuracy. FIGARCH does not systematically dominate simpler GARCH or IGARCH models in this context.
4. **Asymmetry between long and short positions** is pronounced for stock indices but not for exchange rates. This suggests that equity markets react differently to positive and negative shocks and that VaR backtesting must consider both sides of the distribution.
5. **Model selection based solely on AIC/SBC may be misleading** when the objective is risk measurement. Direct evaluation of VaR performance via coverage tests is essential.

In practice, the findings suggest that risk managers should:

- Move beyond simple normal-error models when estimating VaR at extreme quantiles;
- Consider IGARCH or closely related specifications with Student-t errors for 1% and 2.5% VaR;
- Continue to use RiskMetrics or similar EWMA models for 5% VaR where robustness and simplicity are priorities.

Overall, the paper provides strong empirical evidence that the choice of error distribution and volatility dynamics has a substantial impact on VaR estimates and that model comparison should be based on the specific risk management task at hand.

Appendix A: Python Code for the Simulation

Below is the Python code (run in Google Colab) used to generate the illustrative simulation results in Section 7.

```
!pip install arch

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
from arch import arch_model
from scipy.stats import norm, t as student_t

T = 10000
burn = 200
mu = 0.0005
omega = 0.00001
alpha = 0.05
beta = 0.90
nu_true = 5
alpha_VaR = 0.01
np.random.seed(123)

def simulate_garch(normal=True):
    eps = np.zeros(T + burn)
    sigma2 = np.zeros(T + burn)
    sigma2[0] = omega / (1 - alpha - beta)
    for t in range(1, T + burn):
        if normal:
            z = np.random.normal()
        else:
            z_raw = np.random.standard_t(df=nu_true)
            z = z_raw / np.sqrt(nu_true / (nu_true - 2))
        eps[t] = np.sqrt(sigma2[t-1]) * z
        sigma2[t] = omega + alpha * eps[t-1]**2 + beta * sigma2[t-1]
    r = mu + eps
    return r[burn:], np.sqrt(sigma2[burn:])

returns_normal, _ = simulate_garch(normal=True)
returns_t, _ = simulate_garch(normal=False)

model_normal = arch_model(returns_normal, vol='GARCH', p=1, q=1,
                          mean='Constant', dist='normal')
res_normal = model_normal.fit(disp='off')

model_t = arch_model(returns_t, vol='GARCH', p=1, q=1,
                     mean='Constant', dist='t')
res_t = model_t.fit(disp='off')
```

```
sigma_normal_hat = res_normal.conditional_volatility
sigma_t_hat = res_t.conditional_volatility

z_1pct = norm.ppf(alpha_VaR)
nu_est = res_t.params["nu"]
t_1pct_unscaled = student_t.ppf(alpha_VaR, df=nu_est)
t_1pct = t_1pct_unscaled / np.sqrt(nu_est / (nu_est - 2))

VaR_normal = mu + sigma_normal_hat * z_1pct
VaR_t = mu + sigma_t_hat * t_1pct

def empirical_coverage(returns, VaR):
    return np.mean(returns < VaR) * 100

coverage_normal = empirical_coverage(returns_normal, VaR_normal)
coverage_t = empirical_coverage(returns_t, VaR_t)

results = pd.DataFrame({
    "Model": ["GARCH(1,1) Normal", "GARCH(1,1) Student-t"],
    "Target VaR Level (%)": [100*alpha_VaR, 100*alpha_VaR],
    "Empirical Coverage (%)": [coverage_normal, coverage_t]
})
print(results)
```

Reference

So, M. K. P., & Yu, P. L. H. (2006). Empirical analysis of GARCH models in value at risk estimation. *International Financial Markets, Institutions and Money*, 16(2), 180–197.